HAT-P-3b

R-1397

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-3_210629` · created 2026-08-01T03:48:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459394.7981 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.044
Transit depth (Rp/Rs)^2	2.13 +/- 1.29 [%]
Orbital Inclination (inc)	85.1 +/- 1.97
Airmass coefficient 1 (a1)	1.069 +/- 0.016
Airmass coefficient 2 (a2)	-0.053 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#1 - [514, 234]
Optimal Aperture	4.37
Optimal Annulus	7.5
Transit Duration (day)	0.03 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.044 vs 0.1063 (deviation 0.62σ)
Duration fit vs published	0.03 d vs 0.0867 d (deviation 1.96σ)
Mid-time O–C	8.8 min
Depth SNR	2.3
Residual scatter	1.47 %

Light curve

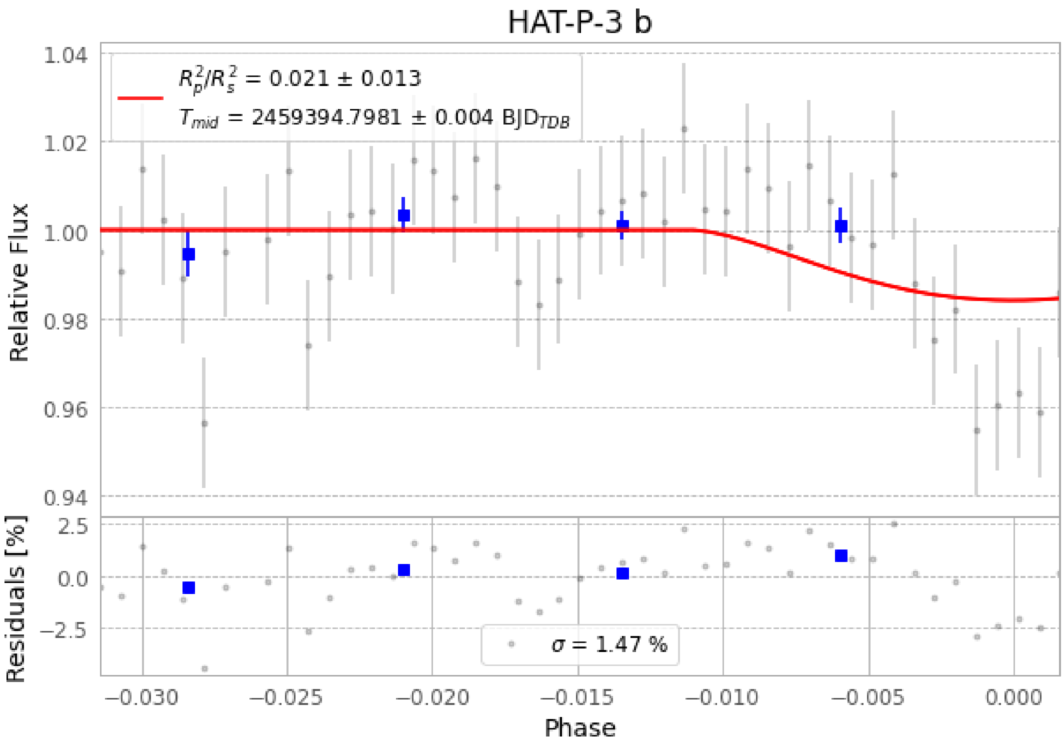

Figure 1 — FinalLightCurve_HAT-P-3 b_2021-06-28.png (SHA-256 33cefb19b477d37c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 252], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c98bd7ff6445be6e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

51 FITS frames (33 MB), `HATP-3210629045712.FITS` → `HATP-3210629072712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210629.log (d1a02db742a8…), FinalLightCurve_HAT-P-3 b_2021-06-28.png (33cefb19b477…), FinalLightCurve_HAT-P-3 b_2021-06-28.csv (de236eb7d9f1…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)